

claude-windows / 42211f80-2169-4025-ad9d-99c1aa29ff70

Metadata

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- Source: `claude-windows`  
- Kind: `claude`  
- Source file: `C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\42211f80-2169-4025-ad9d-99c1aa29ff70\subagents\workflows\wf_6237b110-d58\agent-a847f6ba9f1bee8ad.jsonl`  
- SHA-256: `370e5af8911b95c5ad23cf71300ae72e63ce33e4e0d033384ca2e71372632ef7`  
- Source modified: `2026-06-06T17:39:58+00:00`  
- Imported at: `2026-07-05T16:48:28+00:00`  
- project: `wf_6237b110-d58`  
- session_id: `42211f80-2169-4025-ad9d-99c1aa29ff70`
```

Transcript

1. user (2026-06-06T17:38:57.142Z)

Adversarial Claim Verifier (voter 1/3)

Be SKEPTICAL. Try to REFUTE this claim. ?2/3 refutations kill it.

Research question

Research alternative methods to automatically detect rally start/end segments (temporal segmentation) in racket-sport video ? especially badminton ? for a SELF-HOSTED, LOCAL-FIRST pipeline, and assess whether a cloud-VLM teacher ? local-model distillation is the best approach or whether other methods are competitive or complementary. Keep it focused ("mini" research): prioritize actionable, commercially-licensable, locally-runnable approaches over exotic SOTA needing huge compute or restrictive licenses.

OUR CONTEXT / CONSTRAINTS (ground every recommendation in these):

- Self-hosted on a single consumer GPU (RTX 2070 Super), Windows + WSL2. We want LOCAL-ONLY inference at runtime ? privacy lever: user video never leaves the device. It's a commercial product, so PERMISSIVE / commercial-friendly licenses are REQUIRED (avoid AGPL and non-commercial-only weights/datasets).
- Current substrate: WASB (MIT) shuttle-trajectory detector ? per-frame (x, y, visibility). We derive rally windows via velocity/merge-gap/min-duration thresholds plus a net-crossing gate. We ALSO have a permissive person detector available (torchvision Faster R-CNN + supervision ByteTrack) and the video's AUDIO track.
- Current best approach: use paid Gemini (cloud VLM, full-context) OFFLINE as a TEACHER to label rallies, then DISTILL into a local model (logistic regression on shuttle-trajectory features). Findings so far: threshold-tuning ceilings around F1 ? 0.44 and overfits with few videos; the learned model is data-starved at n=2 labeled videos; WASB candidate RECALL is a real bottleneck (its windows sometimes don't align with true rallies at IoU 0.5).
- Inputs we can exploit locally: video frames, the shuttle trajectory, player bounding boxes / pose (if we add a permissive pose model), and the AUDIO track (shuttle "thwack" hits, crowd noise, umpire/score calls).

SPECIFICALLY COVER:

1. Method families for rally/point segmentation in racket sports (badminton, tennis, table tennis, pickleball): audio-based shot/hit detection; temporal action localization (TAL) deep models; pose/player-stance + court-homography state machines (e.g., "serve-ready stance" gating); classical change-point detection / HMM on the shuttle trajectory; scoreboard / score-overlay OCR for point boundaries; self-supervised / weakly-supervised approaches; and VLM teacher?student distillation.
2. For EACH family: how it works, the labels/data it needs, expected accuracy, local-inference COMPUTE cost, and LICENSE / commercial viability of any models, datasets, or weights involved.
3. Badminton rally-segmentation datasets (e.g., ShuttleSet, TrackNet/TrackNetV2 data) and their licenses / commercial usability.
4. A RANKED recommendation for our local-first, permissive, single-GPU setup: is Gemini-teacher ? local-distillation the best, or are some alternatives competitive or complementary ?

especially AUDIO-based hit detection and pose/court-geometry state machines, which we can run fully locally?

5. Concrete, cheap next experiments we could run with what we already have (shuttle trajectory + person detector + audio).

Claim under review

"Naive audio-only hit detection using raw frame probabilities is poor (F-score ~41-50), but applying the likelihood-ratio confidence measure raises audio-only F-score to roughly 68-77 across four clips ? quantifying how much a simple post-processing trick matters for noisy broadcast audio."

****Source:**** https://www.researchgate.net/publication/261119682_Detection_of_ball_hits_in_a_tennis_game_using_audio_and_visual_information (primary)

****Supporting quote:**** "When using the (absolute) probability of frames, detection performances are quite poor... The use of the likelihood ratio confidence measure significantly reduces the number of false positives in all four video clips."

Checklist

1. Is the claim actually supported by the quote, or is it an overreach/misread?
2. WebSearch for contradicting evidence ? does any credible source dispute or heavily qualify this?
3. Is the source quality sufficient for the claim's strength? (extraordinary claims need primary sources)
4. Is the claim outdated? (check dates ? old claims about fast-moving fields are suspect)
5. Is this a marketing claim / press release / cherry-picked benchmark / forum speculation?

****refuted=true**** if: unsupported by quote / contradicted / low-quality source for strong claim / outdated / marketing fluff.

****refuted=false**** ONLY if: claim is well-supported, current, and source quality matches claim strength.

Default to refuted=true if uncertain.

Structured output only. Evidence MUST be specific.

2. assistant (2026-06-06T17:39:00.878Z)

I'll verify this claim about audio-based hit detection in tennis. Let me start by examining the source and searching for corroborating/contradicting evidence.

3. assistant (2026-06-06T17:39:03.933Z)

Let me research this claim about the tennis ball-hit detection paper and audio detection F-scores.

4. user (2026-06-06T17:39:06.315Z)

The server returned HTTP 403 Forbidden.

The response body was not retrieved. If this URL requires authentication, use an authenticated tool (e.g. `gh` for GitHub, or an MCP-provided fetch tool) instead of WebFetch.

5. user (2026-06-06T17:39:13.454Z)

Web search results for query: "detection of ball hits tennis game audio visual likelihood ratio confidence measure F-score"

Links: [{"title": "Improved Detection of Ball Hit Events in a Tennis Game Using Multimodal", "url": "https://www.isca-archive.org/avsp_2011/huang11_avsp.pdf"}, {"title": "Detection of Ball Hits in a Tennis Game Using Audio and Visual Information", "url": "http://www.apsipa.org/proceedings_2012/papers/316.pdf"}, {"title": "(PDF) Detection of ball hits in a tennis game using audio and visual information", "url": "https://www.researchgate.net/publication/261119682_Detection_of_ball_hits_in_a_tennis_game_using_audio_and_visual_information"}, {"title": "(PDF) Improved detection of ball hit events in a tennis game using"}]

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multimodal information", "url": "https://www.academia.edu/14129696/Improved_detection_of_ball_hit_events_in_a_tennis_game_using_multimodal_information"}, {"title": "Detection of Anomalous Events in a Tennis Game Using Multimodal Information", "url": "http://www.apsipa.org/proceedings_2015/pdf/68.pdf"}, {"title": "ISCA Archive - Improved detection of ball hit events in a tennis game using multimodal information", "url": "https://www.isca-archive.org/avsp_2011/huang11_avsp.html"}, {"title": "Detection of ball hits in a tennis game using audio and visual information | IEEE Conference Publication | IEEE Xplore", "url": "https://ieeexplore.ieee.org/document/6411936/"}, {"title": "Ball Hit Detection in Table Tennis Games Based on Audio Analysis | IEEE Conference Publication | IEEE Xplore", "url": "https://ieeexplore.ieee.org/document/1699506/"}]
```

I found several research papers on ball hit detection in tennis games that discuss audio-visual approaches, likelihood ratio confidence measures, and F-score metrics. Here are the key findings:

Detection Approaches

A novel framework to detect ball hits in a tennis game combines audio and visual information. Single-mode approaches are not very successful: audio detection suffers from interfering noise and acoustic mismatch, while video detection is made difficult by the small size of the ball and the complex background of the surrounding environment.

Confidence Measures

The likelihood of each audio event class for a frame is estimated using Gaussian mixture models of audio events built from the training-data, and the difference between highest log likelihood and the next highest is used as a confidence measure for that frame. The use of the likelihood ratio confidence measure significantly reduces the number of false positives in all four video clips.

Performance Metrics

After taking the inter-ball-hit timing information into account, there are improvements of the F-score values on the first three video clips, though not on the fourth video clip. Combining audio and visual data improves detection metrics across all cases, specifically aiding in accurate identification of volleys and minimizing bounce misrecognitions.

These papers demonstrate that multimodal fusion of audio and visual information with confidence measures based on likelihood ratios provides more robust ball hit detection than single-modality approaches.

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

6. assistant (2026-06-06T17:39:17.373Z)

The primary source returned 403, but I found two directly accessible PDFs from the same research group (APSIPA proceedings and ISCA archive). Let me fetch the actual paper PDF that matches the claim's source, plus the companion paper, to extract exact F-score numbers.

7. user (2026-06-06T17:39:22.411Z)

I cannot find any information about F-scores, precision, recall numbers, audio-only ball-hit detection, or video clip performance metrics in the provided web page content.

The page is from APSIPA (Asia-Pacific Signal and Information Processing Association) and contains organizational information, membership details, conference announcements, and news. It does not include technical research data, detection metrics, or the specific numerical performance measurements you're requesting.

To locate this information, you would need to consult the actual research paper or technical

documentation that reports these audio detection results.

8. user (2026-06-06T17:39:25.664Z)

I apologize, but I cannot extract the specific F-score, precision, and recall metrics you're requesting from this document. The content provided appears to be a PDF file in a compressed/encoded format that contains primarily binary data and embedded font information rather than readable text with performance metrics.

The document structure shows PDF objects (streams, filters, font definitions) but does not display clearly readable performance evaluation numbers for audio-only ball-hit detection across four video clips, nor does it show the specific ranges (41-50 and 68-77) you mentioned.

To obtain the specific metrics you need? F-scores for absolute frame probabilities versus likelihood-ratio confidence measures per video clip? I would need:

- A properly decoded/extracted text version of this PDF, or
- A different document format (plain text, CSV, or HTML) containing the results table

Could you provide the document in an alternate format or share a plaintext excerpt of the results section?

[Binary content (application/pdf, 515.3KB) also saved to C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\42211f80-2169-4025-ad9d-99c1aa29ff70\tool-results\webfetch-1780767075390-n4kl6x.pdf]

9. assistant (2026-06-06T17:39:28.562Z)

The ISCA PDF was downloaded but couldn't be parsed by WebFetch. Let me read the downloaded PDF directly with the Read tool, which handles PDFs.

10. user (2026-06-06T17:39:29.215Z)

PDF file read: C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\42211f80-2169-4025-ad9d-99c1aa29ff70\tool-results\webfetch-1780767075390-n4kl6x.pdf (515.3KB)

11. assistant (2026-06-06T17:39:45.860Z)

I now have the exact data from Table 3 of the paper. Let me verify the claim against the actual numbers.

The claim states:

- "Naive audio-only hit detection using raw frame probabilities is poor (F-score ~41-50)"
- "applying the likelihood-ratio confidence measure raises audio-only F-score to roughly 68-77 across four clips"

Table 3 (audio-only) shows exactly:

****Probability (raw frame probabilities) F-scores:**** VC1=49.56, VC2=49.36, VC3=46.52, VC4=41.23 ? range 41.23?49.56, i.e. "~41-50". ? Exact match.

****Likelihood Ratio (LR) F-scores:**** VC1=72.87, VC2=73.02, VC3=67.81, VC4=77.08 ? range 67.81?77.08, i.e. "roughly 68-77". ? Match (lowest is 67.81, which rounds to 68; highest is 77.08, i.e. ~77).

The claim is precisely supported by the source's own Table 3. The supporting quote ("When using the (absolute) probability of frames, detection performances are quite poor... The use of the likelihood ratio confidence measure significantly reduces the number of false positives in all four video clips") matches the narrative, and the numbers match the table exactly.

Let me run the adversarial checklist:

1. ****Supported by quote/source?**** Yes ? the exact numbers are in Table 3. The "41-50" and "68-77" ranges map directly to the Probability row (41.23?49.56) and LR row (67.81?77.08).

2. ****Contradicting evidence?**** None found. This is a primary peer-reviewed conference paper (AVSP 2011 / the APSIPA 2012 companion). The data is the authors' own reported benchmark.

3. ****Source quality?**** Sufficient ? primary peer-reviewed source from established research groups (UEA + Surrey CVSSP). The claim is not extraordinary; it's a direct restatement of the paper's results table.

4. ****Outdated?**** The paper is from 2011/2012. However, the claim is narrowly about what THIS specific paper reported (a historical/factual claim about a documented result), not a claim about current SOTA. The claim is correctly scoped to that paper's findings, so age does not refute it.

5. ****Marketing/cherry-picked?**** No ? academic conference paper, full per-clip results disclosed (not cherry-picked).

One minor scoping nuance worth noting for honesty: the claim is on a 2011-era GMM/MFCC system on ****tennis broadcast**** audio, not badminton ? and the test setup is somewhat favorable (training/testing within the same tournament-style audio, evaluated against manually-annotated audio ball-hit ranges). But the claim does not overstate generality ? it says "for noisy broadcast audio" and attributes the lift to the likelihood-ratio post-processing trick, which is exactly what the paper concludes. The numbers are not overreached.

The claim survives all refutation attempts. The numbers in the claim match the source's Table 3 essentially exactly.

12. user (2026-06-06T17:39:55.457Z)

Structured output provided successfully

13. assistant (2026-06-06T17:39:58.466Z)

Claim verified as accurate. The exact F-score numbers in the claim (41-50 for raw probability, 68-77 for likelihood-ratio) match Table 3 of the primary source essentially exactly, retrieved from the ISCA archive copy of the paper. Not refuted.